Causality – Francesco Locatello

Experiment Design:

- #1: Give vaccine to children, compare to prev. year. → confirm. bias, bad idea
- #2: ↗ consent → test group
  ↘ non consent → control group } bad, sel. bias
- #3: Population ↗ consent → randomly split ↙ test group || ↘ control group
  ↘ non consent

$\underbrace{\text{Population}}$ → randomized trial, RCT → gold standard

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Causal Inference from observational data (or Discovery)

→ probabilistic inference (causal)

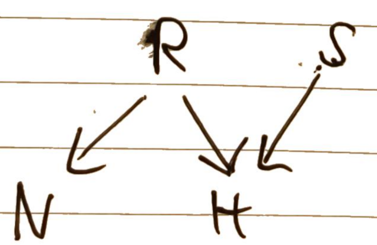
- example:
  - $H=1$: Holmes lawn is wet
  - $R=1$: Rain
  - $S=1$: Sprinkler on

  N: Neighbors lawn wet

$$\Rightarrow P(R=1 \text{ or } S=1 \mid H=1) > P(R=1 \text{ or } S=1)$$

Baysian Networks:

Parent	Child
x	→ y



Parents of (i) in Graph

$$\text{Def: } P(x_1, \dots x_n) = \prod_{i=1}^{N} P(x_i \mid \overbrace{PA(i)})$$

$\Rightarrow \underbrace{P(N, H, R, S)}_{2^4 - 1} = P(H | R, S)\, P(N | R)\, P(R)\, P(S)$

We need DAGs for that!
↳ factorization of joint distribution

DEF: Local Markov Property:

Let $G := (V, E)$ be a DAG, $S \subseteq V$ is arbitr. subset

$$X_S \perp\!\!\!\perp X_{\text{non descendant}}(S) / PA(S) \mid PA(S)$$

↑ "without" (set)

DEF: Collider Node: Non endpoint node in DAG is a collider node if path contains $\rightarrow i \leftarrow$

DEF: d-Seperation: A path P between nodes i and j is Blocked by a set $S \subset V \setminus \{i, j\}$ if

- $\exists$ non-collider $k \in P, k \in S$ $\quad \rightarrow k \rightarrow$, $\leftarrow k \leftarrow$
- $\exists$ collider $k \in P$, $k \notin S$ and descendants $(k) \notin S$ $\quad \leftarrow k \rightarrow$

DEF: if all paths between i, j are blocked by S

$i \perp_d j \mid S$ $\quad$ ($\perp_d$: d-sep),
($\perp\!\!\!\perp_P$: independence)

DEF: Global Markov Property:

P statisfies G.M with respect to DAG G if $A \subset S$, $B \subset S$, $A \cap B = \emptyset$ then

$$A \perp_d B \mid S \Rightarrow X_A \perp\!\!\!\perp_P X_B \mid S$$

$$\begin{tikzcd} 1 \arrow[r] \arrow[dr] & 2 \arrow[d] & \\ 3 \arrow[r] & 4 \arrow[r] & 5 \end{tikzcd}$$

$x_1 \perp\!\!\!\perp x_3$

$x_1 \not\!\perp\!\!\!\perp x_3 \mid x_4$ or x_5

DEF's above are important for Causal Discovery

CAUSAL BAYESIAN NETWORK

Let $G = (V, E)$ (Vertices, Edges) be a DAG and P be the Distribution of X_V. The Pair (G, P) is a CAUSAL DAG MODEL (CAUSAL BAY. Net) if

$W \subset V$ $\quad P(X_V \mid do(X_W = x'_W)) =$

$$= \prod_{i \in V \setminus W} P(x_i \mid PA(i)) \cdot \delta(X_W = x'_W)$$

$$= \begin{cases} \prod_{i \in V \setminus W} P(x_i \mid PA(i)) & \text{if } X_W = x'_W \\ 0 & \text{otherwise} \end{cases}$$

__Example:__ $P(Y = 1 \mid X = 1) > P(Y = 1 \mid X = 0)$

Pass Exam   Attend Lecture

$\Rightarrow P(Y = 1 \mid do(X = 1)) > P(Y = 1 \mid do(X = 0))$

Conditioning: unforced, $do(\cdot)$: forced

Let X has a causal effect on Y if $\exists\, a, b$ s.t.

$P(y \mid do(x = a)) \neq P(y \mid do(x = b))$

$$ATE(x, x') = \mathbb{E}[Y \mid do(X = x)] - \mathbb{E}[Y \mid do(X = x')]$$

confounding
$$\begin{tikzcd} & Z \arrow[dl] \arrow[dr] & \\ X \arrow[rr] & & Y \end{tikzcd}$$
$X, Y \in V$.

The causal effect ~~of~~ between X, Y is confounded if $P(y \mid x) \neq P(y \mid do(x))$

$$P(y \mid do(X=1)) = \sum_z P(y \mid X=1, Z=z) \cdot P(Z=z)$$

Structural Equation Models (SEM)

Let $\mathcal{G}$ be a CAUSAL DAG over $x_1, \dots x_d$.

A SCM/SEM is a collection of structural assignments

$$\{x_j = f_j(PA(j), N_j)\}_{j=1}^{d} \text{ and } P(N_1, \dots N_d) = \prod_i P(N_i)$$

PROP: SCM $\rightarrow$ Unique P

Example:
- $P \in \{bus, subway\}$ (choose transport)
- $T \in \{yes, no\}$ (buy ticket)
- $C \in \{bus, subway\}$ (controller in ...)
- $F \in \mathbb{N}$ $\longrightarrow$ (Fine to pay)

"if bus, always buy ticket, if subway, buy ticket w. $p = 0.8$"

$$\begin{tikzcd} P \arrow[d] \arrow[dr] & C \arrow[d] \\ T \arrow[r] & F \end{tikzcd}$$

$P \sim P_{Bus} = 0.5$ (Bern(0.5))

$T \leftarrow$ bus or $N_T = 1$, $N_T \sim$ Bern(0.8)

$C \sim P_{Bus} = 0.3$, $F = 100 \cdot \delta(P = C)\delta(T = no)$

$\Rightarrow \mathbb{E}[F] \sim$ joint distr., bays. modelling

$\mathbb{E}[F \mid T = no]$

INTERVENTION: $\underbrace{x_j = f_j(PA(j), N_j) \xrightarrow{=} x_j = x_i}_{\text{HARD, PERFECT INTERV.}}$

changing total x_j to x_i

$\underbrace{\text{Set } N_j = N_i \quad do(x_i = N_i) \text{ with } N_i \sim P}_{\text{IMPERFECT INTERVENTION}}$

$do(T = no)$ (Never buy a ticket) | every bayesian is SCM but not vice versa

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COUNTERFACTUAL SCM:

$(S, P(N))$ is an SCM. Given observations

x Counterfactual SCM: $(S, P(N \mid X = x))$

exmpl. $x = (P = bus, C = bus, T = yes, F = 0)$

but what could have happend if T = no?

eg. helpful for policy evaluation

POTENTIAL OUTCOMES: GOLD STANDARD for Obs. Data

$Y_i(X=0)$ and $Y_i(X=1)$ are defined before

the treatment assignment

ATE: $\frac{1}{n} \sum_i Y_i(X=1) - Y_i(X=0)$ | only observe $X=0$ or $X=1$

Day 2 Causality

DEF: GLOBAL MARKOV PROPERTY

$$A \perp_d B \mid S \Rightarrow X_A \perp\!\!\!\perp_P X_B \mid X_S$$

DEF: ADJUSTMENT FORMULA: A set $Z \subset V \setminus \{i, j\}$ is a valid adjustment set if

$$\underbrace{P(x_i \mid do(x_j))}_{\text{causal estimand}} = \underbrace{\int_{x_Z} P(x_i \mid x_j, x_Z)\, P(x_Z)\, dx_Z}_{\text{statistical estimand}}$$

$$\begin{tikzcd} & Z \arrow[dl] \arrow[dr] & \\ x \arrow[rr] & & y \end{tikzcd}$$

DEF: LOCAL MARKOV: $X_j \perp\!\!\!\perp X_{nondescendent(j)} \mid PA(j) \mid X_{PA_j}$

DEF: MARKOV FACTORIZATION:

$$P(x_V) = \prod_i P(x_i \mid PA(i))$$

CAUSAL DISCOVERY

$P(X, Y)$ there is an SCM $Y = f_Y(X, N_Y)$ with $X \perp\!\!\!\perp N_Y$ ... but $x \to y$ or $y \to x$???

FAITHFULNESS: $A \perp_d B \mid S \Leftarrow X_A \perp\!\!\!\perp X_B \mid X_S$

$x_1 \leftarrow \varepsilon_1$

$x_2 \leftarrow 2x_1 + \varepsilon_2$

$x_3 \leftarrow 4x_1 - 2x_2 + \varepsilon_3$

$= \varepsilon_2 + \varepsilon_3$

$$\begin{tikzcd} x_1 \arrow[r] \arrow[dr] & x_2 \arrow[d] \\ & x_3 \end{tikzcd}$$

Questions

- $x_1 \perp\!\!\!\perp x_3$: ✓
- $x_1 \perp_d x_3$: ✗

cond. independence is not d-seperation

Is P FAITHFUL to $\mathcal{G}$? ✗

Is P Markov w.r.t. $\mathcal{G}$? ✓

DEF: MARKOV EQUIVALENT:

TWO DAGs $\mathcal{G}_1, \mathcal{G}_2$ are MARKOV EQUIVALENT if

$$A \perp_d B \mid S \underset{in}{\sim} \mathcal{G}_1 \iff A \perp_d B \mid S \underset{in}{\sim} \mathcal{G}_2$$

PC (Peter & Clark) Algorithm

STEP 1: Find Skeleton (Graph, no directions)

- START FULL GRAPH (any to any)

for $k = 0, \dots, \#\text{Vertices} - 2$

for i, j adjacent in $\mathcal{G}$

Remove edge if $X_i \perp\!\!\!\perp X_j \mid X_S$

where $X_S \subseteq$ adjacent $(i, \mathcal{G})$ or adj$(j, \mathcal{G})$

and $|S| = k$

$$i \not\!\!-\!\!\!- j \underset{\text{Loc. Markov}}{\iff} i \perp_d j \mid PA(i) \text{ or } PA(j) \iff$$

$$\iff i \perp_d j \mid \underbrace{Adj(i) \text{ or } Adj(j)}_{S}$$

$$\underset{\text{Global Markov}}{\iff} X_i \perp\!\!\!\perp X_j \mid X_S$$

STEP 2: Find V-Structure

- $i \rightarrow j \leftarrow k \iff i \not\perp\!\!\!\perp k \mid j$

STEP 3: Orient all other possible edges

↳ no cycles, no new V-structure

ML-based Approaches (Linear)

$x \to y$, $y = f_{lin}(x, N_y) = \alpha x + \underbrace{N_y}$ non gaussian noise

LINGAM ALG.

$X = BX + \mathcal{E}$, $X \in \mathbb{R}^p$, $B \in \mathbb{R}^{p \times p}$ with $\mathrm{diag}(B) = 0$

$\exists$ permutate π s.t. B_π

e.g. $B_\pi = \begin{bmatrix} 0 & 0 & 0 \\ \cdot & 0 & 0 \\ \cdot & \cdot & 0 \end{bmatrix}$, $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \mathcal{E}_1 \\ \mathcal{E}_2 \end{bmatrix}$

STEP 1: $(I - B)X = \mathcal{E} \Rightarrow X = \underbrace{(I-B)^{-1}}_{\text{Matrix?}} \mathcal{E}$

observe

$\mathcal{E}_i \perp\!\!\!\perp \mathcal{E}_j$

$A = (I - B)^{-1}$

A: ICA ~~ICA~~ Indep. Comp. Anal.

not identifiable if gaussian.

Find π of rows s.t. A_π^{-1} is non-zero on diagonal

No TEARS

$X = BX + \mathcal{E} \longrightarrow \min_{B \text{ is DAG}} \|X - BX\|^2 + \lambda \|B\|_1$

„B is DAG" constraint is discrete, hard to optimize, relax by re-writing it continous!

↳ B is DAG $\Rightarrow \mathrm{Tr}\, e^{B \odot B}$ } non convex constraint

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RESIT: Assume $X \to Y$, $y = f_{lin}(x) + \mathcal{E}$

do linear regression $\hat{f}(x) = \hat{y}$, $y - \hat{y} = \underbrace{\mathcal{E} \perp\!\!\!\perp X}_{\text{if true then } x \to y}$

or ML, kernel, NNs...

$\Rightarrow$ expensive, need to regress all variables on all other variables

Non-Linear Methods

$$x_i = \underbrace{f_i}_{\text{non linear}}(PA(j)) + N_i, \quad \underbrace{N_i \sim N(0, \theta_i)}_{\text{strong assumption}}$$

SCORE-BASED

$$\log P(x) = -\frac{1}{2}\sum_i^p \left(\frac{x_i - f_i(PA(i))}{\theta_i}\right)^2 - \frac{1}{2}\sum_i^p \log 2\pi\theta_i^2$$

$$\underbrace{\nabla_x \log P(x)_j}_{S_j(x) =} = \underbrace{\frac{x_j - f_j(PA(j))}{\theta_j}}_{\text{only dep. on } j} + \underbrace{\sum_{i \in \text{Child}_j} \frac{\partial f_i(PA_i)}{\partial x_j} \cdot \frac{x_i - f_i PA}{\theta_i^2}}_{\text{only if children}}$$

$\frac{\partial S_j}{\partial x_j} = \frac{1}{\theta_j^2}$ iff x_j is a leaf $\Rightarrow$ Gives us topological ordering!

Example:

$$S_4 = \frac{x_4 - f_4(PA(4))}{\theta_4^2}$$

$$\begin{tikzcd} & x_1 \arrow[dl] \arrow[dr] & \\ x_2 \arrow[dr] & & x_3 \arrow[dl] \\ & x_4 & \end{tikzcd}$$

θ_{x_4}

$X_2 \rightarrow X_2 \dashrightarrow X_3 \dashrightarrow X_4 \dashrightarrow$

IDENTIFIABILITY MARGIN

$x_j \rightarrow x_i \leftarrow N_i$

$$\mathbb{E}_{p(x_i)}\left[\left(\frac{\partial^2 f_i(PA_i)}{\partial x_j^2}\right)^2\right] \geq \overbrace{c\,\theta_i^2}^{SNR}$$

c is max one can find, like Sign. to Nois Ratio

$\mathrm{Var}\, \frac{\partial S_j}{\partial x_j} \geq C$ if x_j is <u>not</u> a leaf node

$\Rightarrow$ (weak) finite sample guarantees!

Relaxing Gaussian Noise Assumption: SCORE ♡ RESULT

$$S_j(x) = \frac{\partial i}{\partial x_i} \log P(x_i \mid PA(x_i)) + \sum \frac{\partial i}{\partial x_i} f_k (PA_k)$$

$$S = \frac{\partial l}{\partial x_l} \log(P(N_l)) \,,\, N_i = X_i - f(PA_i) \quad \Bigg\| \quad \frac{\partial i}{\partial x_i} \log(P(x_i \mid PA_i))$$

Score of leaf node is non-lin function of its noise

$$R_i = x_i - \underbrace{\mathbb{E}[x_i \mid x_{\setminus \{i\}}]}_{\text{non-lin. regression}}$$

$$= \begin{cases} N_i & \text{if } x_i \text{ is leaf} \\ N_i - \mathbb{E}[N_i \mid x_{\setminus\{i\}}] & \text{otherwise} \end{cases}$$

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everything above from purely observational data

INVARIANT CAUSAL PREDICTIONS

$x_1 \to y \to x_2$ Mode I: $\mathbb{E}[y \mid x_1]$,
Model II: $\mathbb{E}[y \mid x_2]$

CASTING CAUSAL DISCOVERY AS CLASSIFICATION PROBLEM!

CAUSAL REPRESENTATION LEARNING

Causal Point of View on Distribution Shifts

$P(X_1, \dots X_n) = \prod_i P(X_i | PA_i)$

$P(X_1, \dots X_n | do(X_j = x)) = \prod_{i \neq j} P(X_i | PA_i) \delta(X_j = x)$

$\Rightarrow$ Obs. Data + Causal Model = Results from Experim. Studies

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Disentanglement Represent. $\xrightarrow{\text{evolut.}}$ Causal Representations

Structural Equations $\longrightarrow$ Causal Factorization

$S_i = f_i(PA_i, U_i)$ $\quad\quad$ $P(S_1, \dots S_n) = \prod_i P(S_i | PA_i)$

Observations are generated by arbitrary mixing of latents

$X = G(S_1, \dots S_n)$

$\Rightarrow$ Unsupervised Disentanglement Learning is impossible

Key Theorem: For any factorizing distr. over ground truth z there exists a transformation $f(z)$ s.t. ?

z

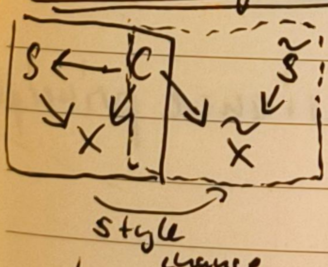
Disantenglement: Each Dimension of the code ~~can~~ only captures single generative factor.

Sparse Mechanism Shifts Assumption: Small distri. changes tend to manifest themselves in a sparse or local .... ??

Weak supervision is enough in theory for learning DisRep known number of fixed factors $|S| = k$, a variability condition on S that z has a continous distribution on that generating function is a diffeormorphism.

?

Causal Representations



soft intervention

$g_{1,2}$

$x \; |||\text{ı} \; \tilde{c}$

$g_{2,1}$

$\tilde{x} \; |||\text{ıı} \; \tilde{c}$

Block Identif.
$\tilde{c} = h(c)$

=> Use self-supervised methods.

See Carey, Aliqyu, Wu, Xiitao: The Fairness Field Guide

CRL-Evaluation: $R^2(z_i, \hat{z}_{A_j}) := \; := \dfrac{\mathbb{V}(\mathbb{E}[z_i \mid \hat{z}_{A_j}])}{\mathbb{V}(z_i)}$

$MCC = \frac{1}{n} \max\limits_{\pi \in \text{perm}} \; ....$?

Causal Estimand that is statistically identified on causal variables is also identifiable from representations

Causal Reps. should make it easier / possible to extract causal information with some downstream estimator.

Conditions for causal validity of downstream estimate: 1. Known $\hat{z}_{A_j} \perp\!\!\!\perp z_i \mid z_{[N] \setminus \{j\}}$

$\hat{z}_i = h(z_j)$ 2. Estimate is invariant to h

Identifiability Algebra

Given two identified blocks of latents:
- Intersection, Complement, Union ?

Key Idea for unified CRL: invariance principle

$v(z_A) = v(\tilde{z}_A) \Leftrightarrow z_A \sim_v \tilde{z}_A$

two vectors have the same projection onto the quotient induced by the equivalence relationship ... ?

z

Dynamical Systems as Causal Models

$$\frac{dx}{dt} = f(x) \quad , \quad x \in \mathbb{R}^d$$ System of coupled differential eq.

$$x(t+dt) = x(t) + dt \cdot f(x(t))$$

ML-Methods: PINNs, SINDy, Neural ODE